Closest Point Algorithm

Finding the Nearest Point

Department: FEST
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1. Problem Statement

We are given one point (P) and a set of other points (S). Our task is to find which point in (S) is nearest to (P).

Given:

$P = (1, -1)$

$S = \{(2, 7), (10, 5), (8, -2), (7, -6), (-3, -5), (-8, 0), (-5, 6)\}$

2. Distance Formula

To find the distance between two points, we use:

$$d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$$

We compare squared distances because we do not need to calculate square roots for every point.

3. Algorithm Steps

1. Take the first point as the nearest point.
2. Calculate its squared distance from (P).
3. Check the remaining points one by one.

4. If a point has a smaller distance, make it the new nearest point.
5. Repeat until all points have been checked.
6. Return the nearest point.

4. Pseudocode

```
None
Algorithm ClosestPoint(P, S)

    best = first point of S

    minDist = squared distance of best from P

    for each remaining point Q in S

        calculate squared distance d

        if d < minDist
            minDist = d
            best = Q
        end if

    end for

    return best
```

5. Solving the Example

Our reference point is $P = (1, -1)$.

The formula becomes:

$d^2 = (x - 1)^2 + (y + 1)^2$

Point	Squared distance
(2,7)	65
(10,5)	117
(8,-2)	50
(7,-6)	61
(-3,-5)	32
(-8,0)	82
(-5,6)	85

The smallest value is **32**, which belongs to the point $(-3, -5)$.

Therefore:

Closest point: $(-3, -5)$

Actual distance:

$d = \sqrt{32} \approx 5.66$

6. Time and Space Complexity

Time Complexity: (O(n))

We check every point once.

Space Complexity: (O(1))

We only store the nearest point and its distance, using a constant amount of extra memory.

7. Conclusion

The Closest Point Algorithm finds the nearest point by comparing distances. In our example, the closest point to ((1,-1)) is ((-3,-5)), and the distance is approximately 5.66 units.